

Integration Project – Model Predictive Control (MPC)

Modeling and Control of a Transport Network

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1 Introduction

This project focuses on the modeling and optimal control of a transport network represented as a directed graph. The objective is to describe the evolution of the volumes stored in the different components of the network and to formulate a Model Predictive Control (MPC) problem in order to optimize the global behavior of the system over a finite time horizon.

First, a mathematical model of the network dynamics is introduced, together with the associated physical constraints. Then, a convex relaxation of the model is proposed in order to make the optimal control problem tractable. Finally, a finite-horizon optimal control problem is formulated.

2 Network Model

The transport network is represented by a directed graph $G = (V, E)$ composed of two types of nodes:

- $\mathcal{T}$: set of **tanks** (reservoirs),
- $\mathcal{X}$: set of **transport cells** (tunnels).

The set of vertices is given by $V = \mathcal{T} \cup \mathcal{X}$, with $\mathcal{T} \cap \mathcal{X} = \emptyset$.

Time is discretized with a constant sampling period $h > 0$:

$$t_k = kh, \quad k = 0, 1, \dots, K.$$

2.1 State and control variables

For each discrete time step k , the following variables are defined:

- $x_T(k)$: water volume stored in tank $T \in \mathcal{T}$,
- $x_i(k)$: water volume stored in cell $i \in \mathcal{X}$,
- $u_T(k) \in [0, 1]$: valve opening control associated with tank T ,
- $f_{ij}(k) \geq 0$: flow from node i to node j ,
- $\mu_i(k)$: outflow leaving the network from terminal cells.

2.2 Demand and supply functions

For each transport cell $i \in \mathcal{X}$, demand and supply functions are introduced.

Demand function The demand function is defined as:

$$d_i(x_i(k)) = \gamma_i \min \left(\frac{v_i}{l_i} x_i(k), C_i \right),$$

where v_i denotes the velocity, l_i the length of the cell, C_i the maximum capacity, and $\gamma_i \in [0, 1]$ is a control parameter used to regulate the flow.

Supply function The supply function is given by:

$$s_i(x_i(k)) = \max (c_i - w_i x_i(k), 0),$$

where c_i is a capacity constant and w_i is a saturation coefficient.

2.3 Flow definitions

Cell-to-cell flow For two transport cells $i, j \in \mathcal{X}$, the flow is defined as:

$$f_{ij}(k) = \min(d_i(x_i(k)), s_j(x_j(k))).$$

Many cells to a single cell flow We have $m \in \mathcal{X}$ cells that all lead to a single cell $j \in \mathcal{X}$. The flow is defined as :

$$\forall i \in [1, m], f_{ij} = \min(\beta d_i(x_i(k)), s_j(x_j(k)))$$

Tank-to-cell flow For a tank $T \in \mathcal{T}$ and a cell $j \in \mathcal{X}$:

$$f_{Tj}(k) = \min(u_T(k) x_T(k), s_j(x_j(k))).$$

Outflow from the network If a cell i has no successor, the outflow leaving the network is:

$$\mu_i(k) = d_i(x_i(k)).$$

No return to tanks Flows from transport cells back to tanks are forbidden:

$$f_{jT}(k) = 0, \quad \forall j \in \mathcal{X}, \forall T \in \mathcal{T}.$$

2.4 Discrete-time system dynamics

Tank dynamics The evolution of the tank volumes is described by:

$$x_T(k+1) = x_T(k) - h \sum_{j: T \rightarrow j} f_{Tj}(k).$$

Cell dynamics The dynamics of the transport cells are given by:

$$x_i(k+1) = x_i(k) + h \left(\sum_{j: j \rightarrow i} f_{ji}(k) - \sum_{m: i \rightarrow m} f_{im}(k) - \mu_i(k) \right), \quad i \in \mathcal{X}.$$

2.5 Physical constraints

The system is subject to the following constraints:

$$\begin{cases} x_i(k) \geq 0, & x_T(k) \geq 0, \\ f_{ij}(k) \geq 0, \\ 0 \leq u_T(k) \leq x_T(k). \end{cases}$$

When we get $m \in \mathcal{X}$ cells to a single cell $j \in \mathcal{X}$, we add these constraints :

$$\sum_{i=1}^m f_{ij} \leq s_j$$

3 Open-Loop Analysis of a Simple Network

3.1 Network Description: Line Graph — Two Tanks and Six Cells

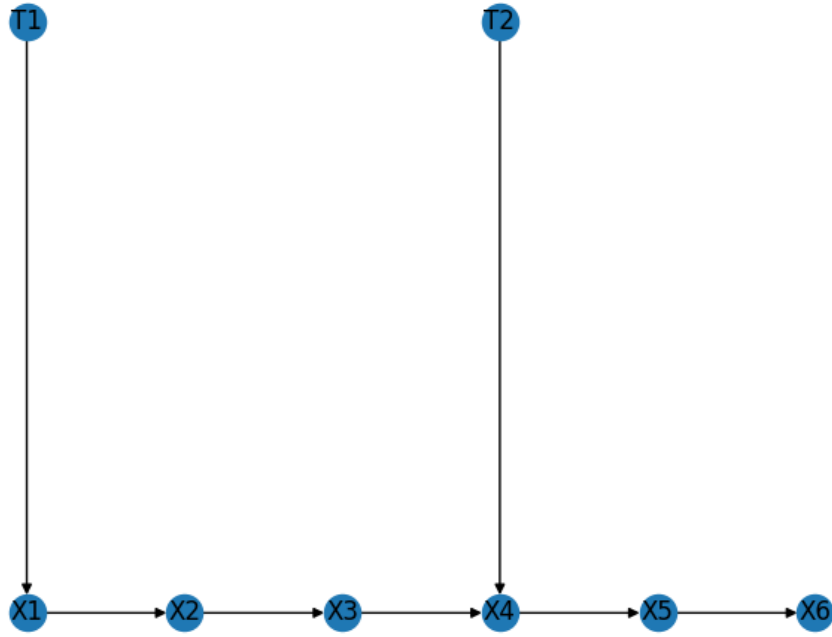


Figure 1: Network with two Tanks and six Cells

First we modeled a simple line graph network with 6 cells and two tanks plugged respectively on the first and the fourth cell as we can see in figure 1. Each cell have a 500 meters length and a capacity of 1000 vehicles/h. Each vehicle move at 70 km/h in the cells. The two tanks are initially filled with 100 vehicles.

3.2 Open-Loop Simulation Setup and Results

To empty the tanks, we will try two naive open-loop strategies. First we will try to empty both tanks at the same time and then we will totally empty one tank and then proceed to empty the other one (figure 2).

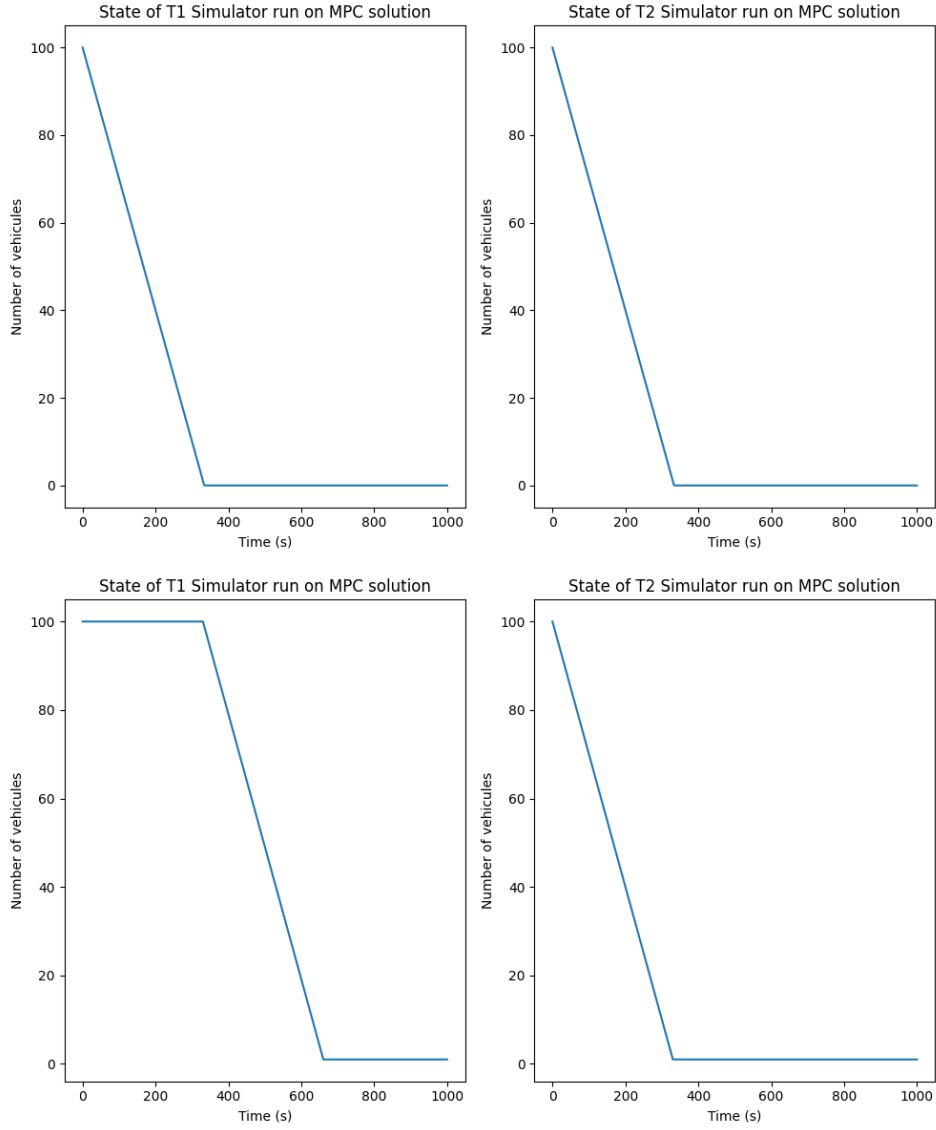


Figure 2: Different ways to empty the tanks : both emptying at the same time or totally empty one and then the other one

When the two tanks are being emptied at the same time (figure 3), there some stress on cell X4 (it peaks at 85 vehicles around 400s). This can be explained by the addition of vehicles coming from the first and second tanks, which accumulate at cell 4. When emptying the tanks one after the other (figure 4) (first tank 2, then tank 1), we notice that we never exceed twenty vehicles, regardless of the cell. Even if it takes a little longer in this configuration, we notice that there is less stress on the cells.

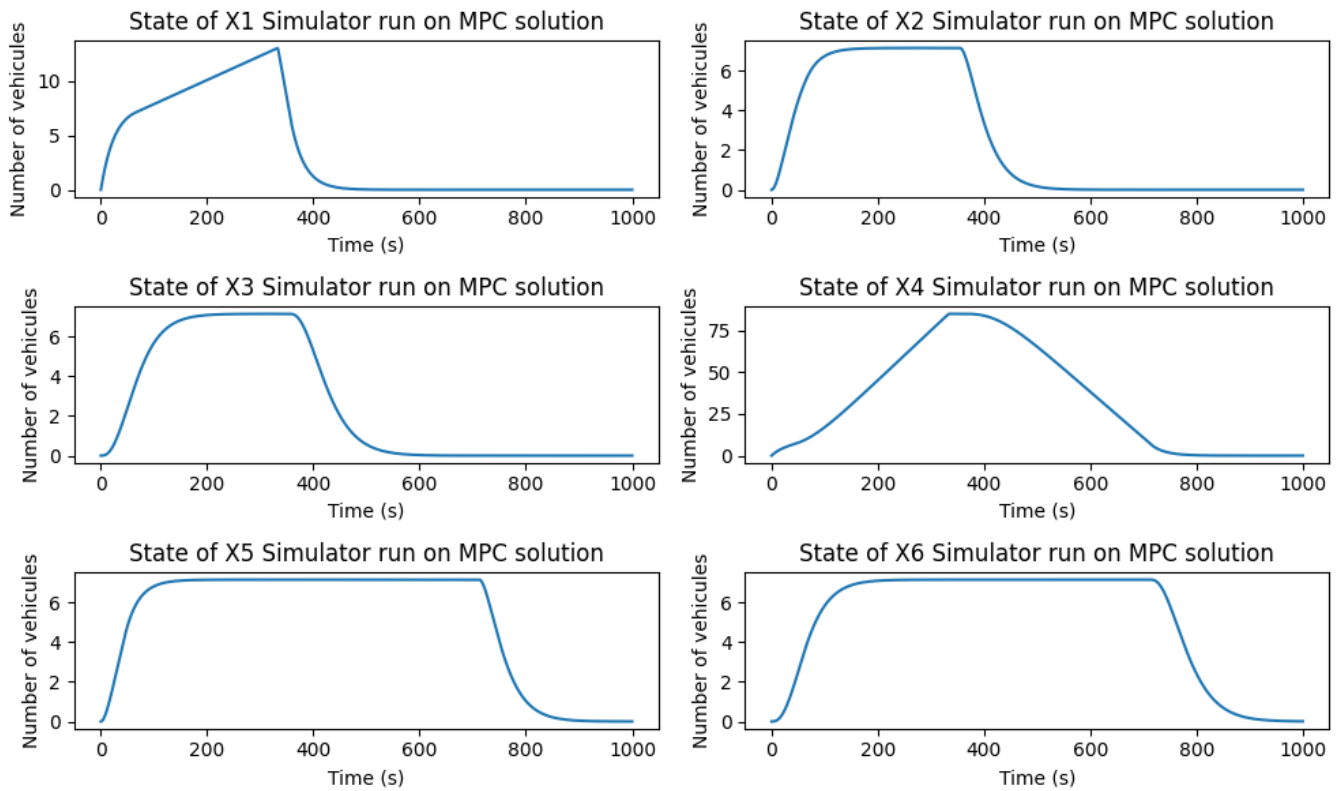


Figure 3: Results when both tanks are emptied at the same time

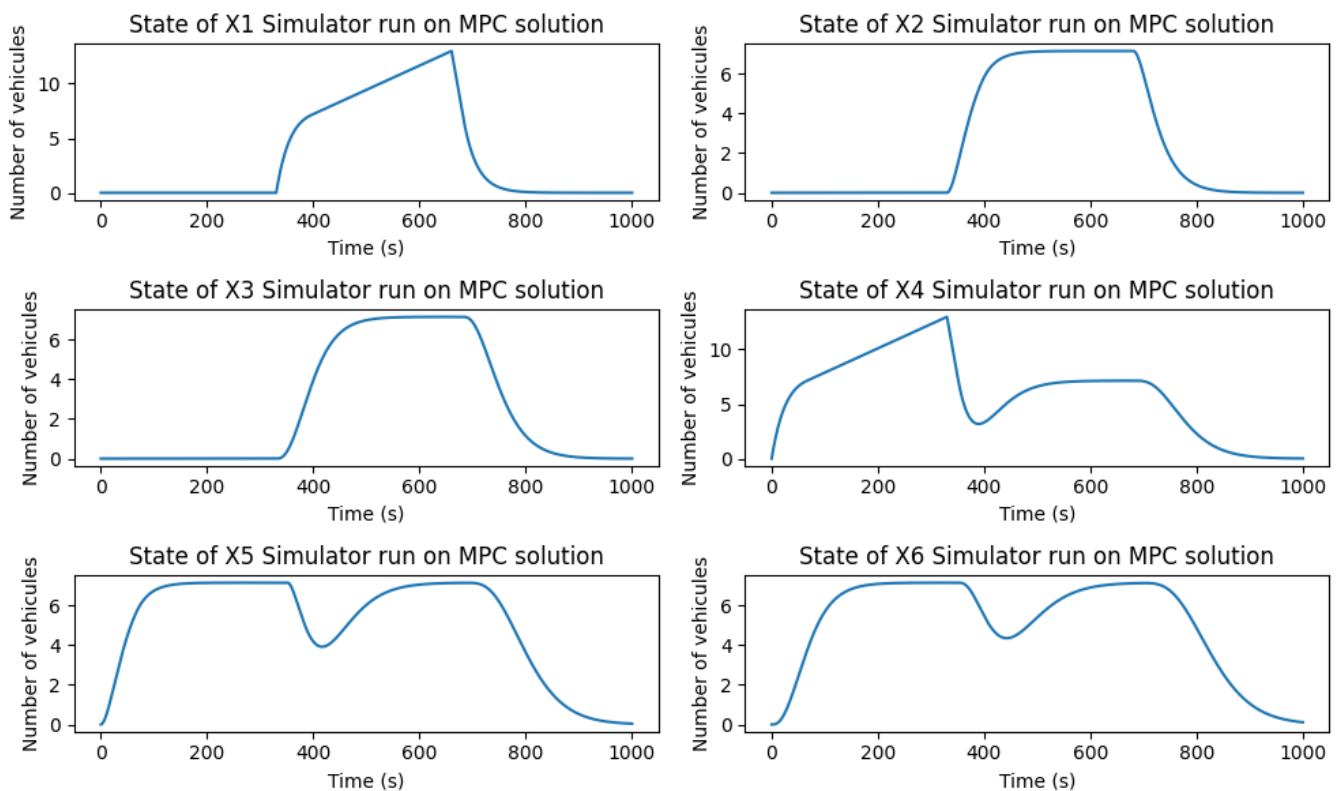


Figure 4: results when the tanks are emptied one after the other

3.3 Discussion

4 Model Predictive Control Formulation

4.1 Convex Relaxation of the Flow Model

To obtain a convex formulation, the flows are treated as independent decision variables and the $\min(\cdot)$ operators are replaced by inequality constraints:

$$\begin{aligned} 0 &\leq \bar{f}_{ij}(k) \leq d_i(x_i(k)), \\ \sum_{j:j \rightarrow i} \bar{f}_{ji}(k) &\leq s_i(x_i(k)). \end{aligned}$$

4.2 MPC Problem Statement : Finite-horizon optimal control problem

A prediction horizon of length K is considered, together with the following cost function:

$$J = \sum_{k=0}^K \left(\sum_{i \in \mathcal{X}} x_i(k) + \sum_{T \in \mathcal{T}} x_T(k) \right).$$

The objective is to minimize J subject to:

- control constraints:

$$0 \leq u_T(k) \leq x_T(k), \quad \gamma_i \in [0, 1],$$

- flow constraints:

$$\begin{aligned} \sum_{m:i \rightarrow m} \bar{f}_{im}(k) &\leq d_i(x_i(k)), \\ \sum_{j:j \rightarrow i} \bar{f}_{ji}(k) &\leq s_i(x_i(k)), \\ \sum_{j:T \rightarrow j} f_{Tj}(k) &\leq u_T(k) x_T(k), \end{aligned}$$

- system dynamics,
- no return flows to tanks,
- given initial condition $x(0)$.

4.3 Numerical Implementation of the MPC Controller

5 Coupling the MPC Controller with the Simulator

5.1 Simulator Description

5.2 Closed-Loop Control Architecture

The control is done following these steps :

- Formulate the problem statement (cost function, constraints)
- Solve the optimization problem
- Apply the first half solutions of the optimization problem
- Go again with the previous steps until we get the last iteration

5.3 Closed-Loop Simulation Results

This part is fed with both the MPC results and the simulator outputs based on the MPC solution. The MPC is run over a finite horizon of 200 for 2000 iterations.

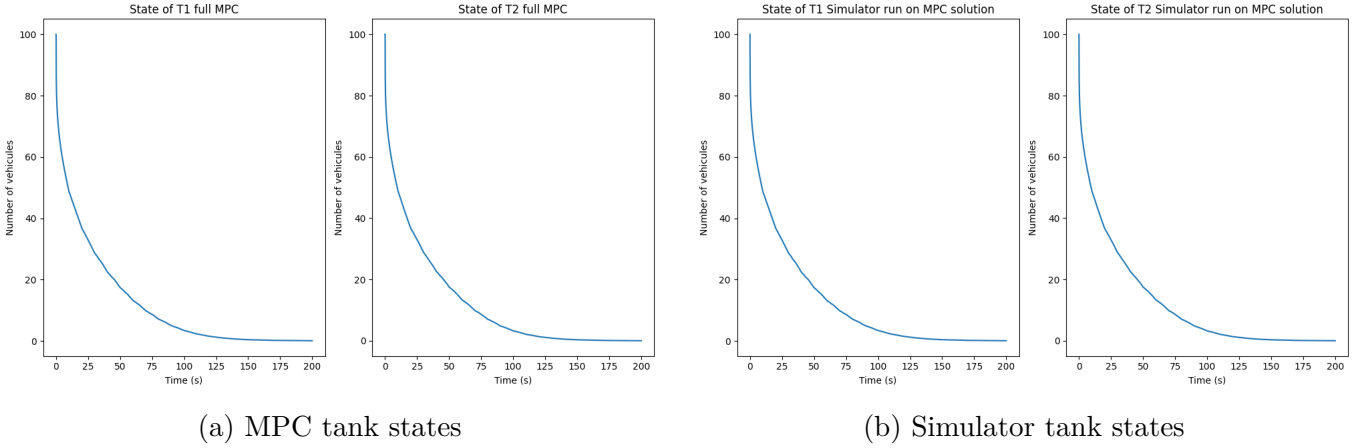


Figure 5: MPC and simulator tank states

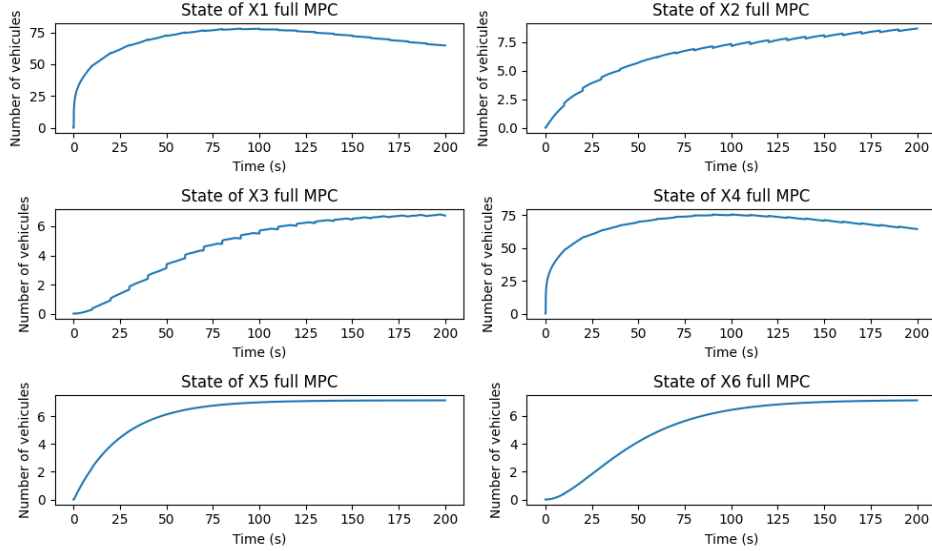


Figure 6: MPC cells states

6 Results and Performance Analysis

6.1 Comparison Between Open-Loop and Closed-Loop Behavior

Table 1 compares the time performance of different control strategies. The results show that Model Predictive Control (MPC) with a prediction horizon of 800 and the optimization-based approach lead to very similar outcomes, with times of 733 s and 732 s respectively, indicating nearly identical closed-loop behavior. In contrast, open-loop strategies result in significantly longer times. When the two tanks empty sequentially, the time increases to 902 s, while it reaches 836 s when both tanks empty simultaneously. These results clearly highlight the advantage of closed-loop control strategies, which provide a more efficient dynamic management of the system and significantly reduce the time required to reach the condition where only one vehicle remains in zone X6.

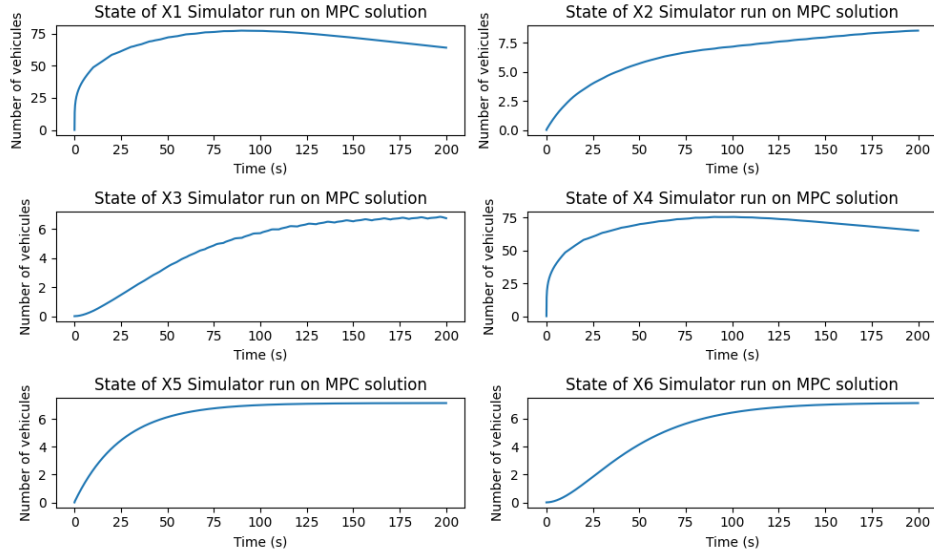


Figure 7: Simulator cells states

Used Method	Time (in seconds) when there are only one vehicle left in X6
MPC with horizon 800	733
Optimization problem	732
Open-loop when the two tanks empty one after the other	902
Open-loop when both tanks empty at the same time	836

Table 1: Comparison Between Open-Loop and Closed-Loop Behavior

6.2 Impact of MPC on Congestion Reduction

6.3 Discussion on Model-Simulator Mismatch

7 Conclusion and Perspectives

- Summary of the main results
- Limitations of the proposed approach
- Perspectives and future work

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